# University of Cape Coast

**Department of Mathematics & Statistics**

**End of First Semester Examination (2015/2016)**

**MAT 101: Algebra and Trigonometry**

2 HOURS

Answer all questions in Section A and any two questions from Section B.

## SECTION A

*Answer all questions in this section.*
*Four options have been provided for each question. Shade on your answer sheet the letter that bears the same answer as the one you have chosen.*

**1.** Find the difference $d$ of an A.P for which $a_1 = 9$ and $S_5 = 15$.

**A.** $-5$ **B.** $-4$ **C.** $-3$ **D.** $2$

**2.** If 6 times the $6^{th}$ term of an A.P is equal to 9 times the $9^{th}$ term, find the $15^{th}$ term.

**A.** 0 **B.** 1 **C.** 2 **D.** 3

**3.** Form the polynomial equation whose zeros are $2+\frac{1}{\sqrt{2}}, 2-\frac{1}{\sqrt{2}}$.

**A.** $x^2-3x-1$ **B.** $2x^2-8x+7$

**C.** $2x^2+x-2$ **D.** $x^2+5x+1$

**4.** In triangle $ABC$, right-angled at $B$, if $\tan A = \frac{1}{\sqrt{3}}$, find the value of $\sin A\cos C + \cos A\sin C$.

**A.** 0 **B.** 1 **C.** 2 **D.** 3

**5.** Which of the following is false?

**A.** The value of $\sin\theta$ increases as $\theta$ increases

**B.** The value of $\cos\theta$ increases as $\theta$ increases

**C.** $\cot A$ is not defined for $A = 0^\circ$

**D.** $\sin(A+B) = \sin A\cos B + \cos A\sin B$

**6.** If $\tan 2A = \cot(A-18^\circ)$, where $2A$ is an acute angle, find the value of $A$.

**A.** $18^\circ$ **B.** $72^\circ$ **C.** $36^\circ$ **D.** $90^\circ$

Given that $\tan(A+B)=\sqrt{3}$ and $\tan(A-B)=\frac{1}{\sqrt{3}}$;

$0^\circ < A+B < 90^\circ$, $A > B$.

Use it to answer Questions **7** and **8**.

**7.** Find $A$.

**A.** $5^\circ$ **B.** $15^\circ$ **C.** $30^\circ$ **D.** $45^\circ$

**8.** Find $B$.

**A.** $5^\circ$ **B.** $15^\circ$ **C.** $30^\circ$ **D.** $45^\circ$

A function $f(x)$ is given by $f(x)=\frac{2x}{x^2+1}$.

Use it to answer Questions **9** to **12**.

**9.** Find the domain of $f(x)$.

**A.** $(-1, 1)$ **B.** $(-\infty,-1)$

**C.** $(-1, 0)$ **D.** $(-\infty, \infty)$

**10.** Find the range of $f(x)$.

**A.** $[-1, 1]$ **B.** $(-1, 1]$

**C.** $[-1, 1)$ **D.** $(-1, 1)$

**11.** Find the maximum turning point.

**A.** $(2, 1)$ **B.** $(-1, 1)$

**C.** $(1, 1)$ **D.** $(-1, -1)$

**12.** For what value of $x$ is $f(x)$ undefined?

**A.** None **B.** $x=-1$

**C.** $x=0$ **D.** $x=1$

13. Simplify $\dfrac{1+\tan^2 A}{1+\cot^2 A}$.

**A.** $\sec^2 A$ **B.** $\sin^2 A$ **C.** $\cot^2 A$ **D.** $\tan^2 A$

14. Factorize the expression $2x^3+3x^2-32x+15$.

**A.** $(x-2)(x-3)(2x-3)$ **B.** $(x+3)(x+5)(2x+1)$

**C.** $(x-3)(x-2)(2x+1)$ **D.** $(x-3)(x+5)(2x-1)$

15. Find the values of $x$ for which the expression $\dfrac{2x+5}{x^2-x-6}$ does not exist.

**A.** $3, -2$ **B.** $-3, 1$ **C.** $3, 2$ **D.** $3, -1$

16. Find the limit of the function $f(x)=\dfrac{2x}{1+x^2}$ as $x \to \infty$.

**A.** $-1$ **B.** $\dfrac{1}{x}$ **C.** $0$ **D.** $2$

The roots of the equation $3x^2+4x-5=0$ are $\alpha$, $\beta$.

Use it to answer Questions **17** to **19**.

17. Find the value of $\dfrac{1}{\alpha}+\dfrac{1}{\beta}$.

**A.** $\dfrac{4}{5}$ **B.** $-\dfrac{4}{3}$ **C.** $2$ **D.** $-\dfrac{4}{5}$

18. Find the value of $\alpha^2+\beta^2$.

**A.** $\dfrac{16}{9}$ **B.** $\dfrac{10}{3}$ **C.** $\dfrac{46}{9}$ **D.** $-\dfrac{4}{3}$

**19.** Find the value of $\frac{\alpha}{\beta}+\frac{\beta}{\alpha}$.

**A.** $\frac{10}{3}$ **B.** $-\frac{46}{15}$ **C.** $\frac{16}{9}$ **D.** $-\frac{4}{3}$

Given that $P(x)=x^5-4x^3+2x+3$.

Use it to answer Questions **20** and **21**.

**20.** Find the remainder when $P(x)$ is divided by $(x-1)$.

**A.** 1 **B.** 2 **C.** 3 **D.** 4

**21.** Find the remainder when $P(x)$ is divided by $(x+2)$.

**A.** 0 **B.** $-1$ **C.** $-2$ **D.** 3

When the expression $x^5+4x^2+ax+b$ is divided by $x^2-1$, the remainder is $2x+3$.

Use it to answer Questions **22** and **23**.

**22.** Find the value of $a$.

**A.** $-1$ **B.** 0 **C.** 1 **D.** 2

**23.** Find the value of $b$.

**A.** $-1$ **B.** 0 **C.** 1 **D.** 2

**24.** If $\mathbf{A}=\begin{pmatrix}1 & 2\\ 3 & 4\\ 5 & 6\end{pmatrix}$ and $\mathbf{B}=\begin{pmatrix}7\\ 8\end{pmatrix}$, find $\mathbf{AB}$.

**A.** $(23 \quad 53 \quad 83)$ **B.** $(23 \quad 83 \quad 53)'$

**C.** $(53 \quad 83 \quad 23)$ **D.** $(23 \quad 53 \quad 83)'$

**25.** Find the values of $x$ for which the matrix $\begin{pmatrix} x-2 & 1 \\ 2 & x-3 \end{pmatrix}$ has no inverse.

**A.** $-1, 2$ **B.** $3, 4$ **C.** $1, 4$ **D.** $-1, -2$

**26.** State which of the following matrices is non-singular.

**A.** $\begin{pmatrix} 3 & -2 \\ 9 & 6 \end{pmatrix}$ **B.** $\begin{pmatrix} 3 & 2 \\ 9 & 6 \end{pmatrix}$

**C.** $\begin{pmatrix} 34 & 119 \\ 26 & 91 \end{pmatrix}$ **D.** $\begin{pmatrix} x & x^2 \\ x^2 & x^3 \end{pmatrix}$

**27.** Find the inverse of the matrix $\begin{pmatrix} x & -1 \\ 1 & x \end{pmatrix}$.

**A.** $\frac{1}{2x-1}\begin{pmatrix} x & 1 \\ -1 & x \end{pmatrix}$ **B.** $\frac{1}{x^2+1}\begin{pmatrix} x & 1 \\ -1 & x \end{pmatrix}$

**C.** $\frac{1}{x^2-1}\begin{pmatrix} x & -1 \\ 1 & x \end{pmatrix}$ **D.** $\frac{1}{x^2-1}\begin{pmatrix} x & 1 \\ -1 & x \end{pmatrix}$

**28.** Evaluate the expression $\begin{pmatrix} n \\ n-1 \end{pmatrix}$.

**A.** $n$ **B.** $n-1$ **C.** $n(n-1)$ **D.** $n!$

**29.** Find the sum of the first twenty terms of the A.P. $-4-1+2+\cdots$.

**A.** 200 **B.** 249 **C.** 490 **D.** 530

**30.** Given that $\sin(x+45^o) = 2\cos(x+45^o)$, find the value of $\tan x$.

**A.** $-\frac{2}{3}$ **B.** $-\frac{1}{3}$ **C.** $\frac{2}{3}$ **D.** $\frac{1}{3}$

## SECTION B

*Answer **two** questions only from this section*
*All questions carry equal marks*

**1.** (a) A function $f(x)$ is given by

$$f(x) = \frac{3x-9}{x^2-x-2}.$$

Examine the behaviour of the function by finding all possible asymptotes and hence, sketch its graph.

Show also that $f(x)$ cannot lie between two certain values.

(b) If $\alpha$, $\beta$ and $\gamma$ are the roots of $x^3 - 4x^2 - 5x - 1 = 0$, without solving the equation, find the equation whose roots are $\alpha^2$, $\beta^2$ and $\gamma^2$.

**2.** (a) Find the characteristic equation of the matrix

$$\mathbf{P} = \begin{pmatrix} 0 & -1 & 1 \\ 2 & -1 & 2 \\ 1 & 0 & 1 \end{pmatrix}.$$

Hence, find the inverse of $\mathbf{P}$.

(b) Use matrix method to solve the linear equations

$$-y + z = 2$$
$$2x - y + 2z = 5$$
$$x + z = 4$$

**3.** (a) Express $g(x) = \sqrt{3}\sin\theta - \cos\theta$ in the form $R\sin(\theta - \alpha)$ where $R > 0$.

(i) Determine the maximum and minimum values of $g$ and the corresponding values $\theta$.

(ii) Solve the equation for $4\sin\theta\cos\theta = \sqrt{3}\sin\theta - \cos\theta$, $0^0 \le \theta \le 360^0$.

(b) Show that if $x$ is small enough to neglect its cube and higher powers,

$$\sqrt{\frac{1-x}{1+x}} = 1 - x + \frac{x^2}{2}.$$

**4.** (a) A sequence is defined by the statements

$$u_1 = 1, \quad u_{n+1} = u_n + \left(\tfrac{1}{4}\right)^n, \ \forall n \ge 1.$$

(i) Write down $u_3$ and $u_4$.

(ii) Find an expression for $u_n$ in terms of $n$ only.

(iii) Evaluate $\lim_{n\to\infty} u_n$.

(b) Solve the simultaneous equation

$$\log_y x - 4\log_x y = 0$$
$$\log_{10} x + \log_{10} y = 1$$

*End*